

Fig. 9(a) shows the block schematic representation of command guidance. Fig. 9(b) shows the deployment scenario.

Wire-guided missiles are an example of command guidance. Here, command signals are sent to the missile through a conventional wire or a fibre-optic cable that actually reels out from the rear of the missile up to the launch platform. The missile trajectory in this case is controlled with the help of command signals transmitted via a wired link rather than a radio link. These missiles are commonly used for short-range anti-tank operations launched from either land based platforms or helicopters. In many cases, even torpedoes fired from submarines use wire guidance.

TOW is a popular example of a wire-guided missile. Manufactured by Hughes Aircraft Company, it is primarily used in anti-tank warfare and is a command to line-of-sight weapon. Current versions are capable of penetrating 76.2cm (30 inches) of armour at a maximum range of three kilometres. It can be fired from a vehicular platform, a helicopter and even by infantrymen using a tripod stand. Fig. 10 shows TOW missile being launched from a land platform.

Command guidance can be classified as command line-of-sight (CLOS) and command off line-of-sight (COLOS) guidance.

CLOS systems are further subdivided into four groups. First is manual command to line-of-sight (MCLOS), where target tracking, missile tracking and control functions are all performed manually. Second type is semi-manual command to line-of-sight (SMCLOS). Here, target tracking is automatic but missile tracking and control functions are performed manually.

The third category is called semi-automatic command to line-of-sight (SACLOS). Here, target tracking is manual and missile tracking and control functions are automatic. SACLOS is the most common form of guidance in use against ground



Fig. 10: TOW missile system

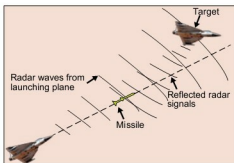


Fig. 11: Semi-active homing guidance



Fig. 12: DRDO-Astra BVRAAM air-to-air missile

targets such as bunkers and tanks. Hellfire from Lockheed Martin is a helicopter-launched fire-and-forget anti-armour air-to-ground weapon of SACLOS category.

First three generations of the weapon use laser seeker, while the fourth generation uses radar seeker. The fourth sub group is known as automatic command to line-of-sight (ACLOS), where all three functions are automatic.

COLOS system, unlike CLOS system, does not depend on angular coordinates of the missile and the target. The guidance system ensures missile interception of the target by locating both missile and target in space for which distance coordinate is needed. This can be possible only if both missile tracker and target tracker were active. In the case of COLOS system, missile and target tracker can be oriented in different directions.

Homing guidance. Homing guidance is most commonly used in

surface-to-air and air-to-air guided weapons. It is further sub-divided into four groups: semi-active homing, active homing, passive homing and track-via-missile homing (also known as re-transmission homing).

In the case of semi-active homing guidance, the target is illuminated by an external source, which could be a radar or laser. The electromagnetic energy reflected by the target is intercepted by the seeker head of the guided weapon. An onboard computer processes the intercepted signal and determines the target's relative trajectory. It sends appropriate command signals to the control surfaces of the weapon to make it intercept the target.

Fig. 11 illustrates the concept of semi-active homing guidance in the case of an air-to-air missile. Semi-active homing is similar to command guidance, except for the fact that in the case of former, the command computer is onboard the weapon. The type of seeker head, whether it is radar seeker or laser seeker, depends on the type of external source designating the target. Both radar as well as laser-guided semi-active homing weapons are in use. Sparrow air-to-air missile and laser-guided weapons of the Paveway family are examples of semi-active homing guidance.

In the case of active homing guidance, the source of target designation is also onboard the weapon, with the result that this methodology does not require an external source. These features put it in the category of fire-and-forget missiles as the launch platform does not need to continue to illuminate the target after the missile has been launched. Active homing guidance weapons are usually radar-guided.

Also, in the case of active homing guidance, transmitted and reflected waves are at the same angle with respect to the line-of-sight between the target and the missile. This is different from semi-active homing mechanism in which transmitted and reflected waves are at an angle. It is because of this reason that semi-active and active